

Dr. Aleksandra Ivanova (formerly Nikonenko)

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Profile

Computational chemist with strong experience in **computer-aided drug design**, **molecular modeling**, **cheminformatics**, and **scientific software development**. My work combines molecular docking, molecular dynamics simulations, binding free-energy analysis, QSAR, *de novo* molecular design, Python-based workflow automation, and HPC execution for small-molecule drug discovery. **Lead developer** of [StreamMD](https://github.com/aleksandra-ivanova/StreamMD) (★ 100+ 📄 20+), an open-source toolkit for automated setup, execution, and analysis of molecular dynamics simulations at scale.

Skills

Programming	Python, Bash, R, Cython, SQL, $\LaTeX$
Drug discovery/CADD	Molecular docking, molecular dynamics simulations, binding free-energy calculations (GBSA/PBSA/FEP/ABFE), QSAR <i>de novo</i> design, protein–ligand interaction analysis, automated preprocessing and quality control of chemical datasets
Cheminformatics	RDKit, MDAnalysis, ProLIF, CReM, GROMACS, OpenMM, Amber, AutoDock Vina, Gnina, Glide, Maestro, gmx_MMPBSA, A3FE, ChemFP, Gaussian, MCPB.py, PyMOL, Chimera
Python ecosystem	scikit-learn, pandas, NumPy, Matplotlib, Plotly, Streamlit
High-performance computing	Dask, PBS, SLURM, multiprocessing, parallel, code performance profiling
Software development	Git/GitHub, package distribution (PyPI, conda-forge), environment management (conda, venv), containerization (Docker, Apptainer), REST API design, construction of reproducible workflows, unit testing, integration testing
LLMs/AI agents	Development of agent-driven scientific workflows; hands-on experience with hosted and local/open-source LLMs, prompt design, tool integration, and structured agent–tool communication

Experience

2026–present	Researcher Institute of Molecular and Translational Medicine, Palacký University, Czech Republic Conduct computational drug discovery research involving molecular dynamics simulations, molecular docking, binding free-energy analysis, QSAR modeling, and <i>de novo</i> molecular design. Develop Python-based workflows and open-source tools for reproducible molecular modeling, including <i>StreamMD</i> and <i>CReM-opt</i> .
Jun–Dec 2022	Computer Scientist (part-time) Al ffinity s.r.o, Czech Republic Curated and validated protein structure (PDB) datasets for MD and ML pipelines; performed molecular docking using Glide.
2017–2019	Laboratory Assistant Laboratory of Chemistry and Biochemistry, Volgograd State University, Russia Provided laboratory assistance in chemistry and biochemistry, including sample preparation, handling laboratory equipment, maintaining laboratory materials, and supporting routine experimental procedures.

Education

2019–2026	Ph.D., Molecular and Translational Medicine — Computational Chemistry Palacký University, Czech Republic Thesis: <i>From molecular conformation to biological activity: computational methods for the design of small molecules</i> . Supervisor: Dr. Pavel Polishchuk
2014–2019	Specialist (Honours), Bioengineering & Bioinformatics Volgograd State University, Russia

Projects

Lead Developer

CReM-agent

Developed an autonomous, agent-driven molecular optimization framework powered by CReM and molecular docking. *Manuscript in preparation*.

StreaMD ★ 100+ 📄 20+ [GitHub](#)

Developed an open-source toolkit for automated setup, execution, and analysis of high-throughput molecular dynamics simulations. Published in *J. Cheminformatics* (2024). *First-author publication*.

CReM-opt [GitHub](#)

Developed a docking-guided molecular optimization framework combining CReM fragment replacement, genetic algorithm, an AI agent, and atom-wise docking score attribution. Presented at *RDKit UGM 2025*. Poster Award at *12th ICCS 2022, Netherlands*. *Manuscript in preparation*.

MIL-based QSAR Modeling

Developed a conformer-based multi-instance learning approach for molecular activity prediction. Automated the collection of activity-cliff datasets (stereoisomers with differential biological activity). *First-author publication*. Contributed to a related study through dataset curation and molecular docking.

ChEMBL Datasets Collection [GitHub](#)

Developed an automated pipeline for retrieving and preprocessing bioactivity datasets from ChEMBL, tailored to specific activity types (e.g., agonists, antagonists, inverse agonists).

HPC Stats Scripts [GitHub](#)

Developed utilities for monitoring SLURM/PBS job efficiency and CPU/GPU/memory usage in HPC environments.

Selected Research Contributions

CACHE Challenge #1 — 3rd Place Team WDR Domain of LRRK2

Developed and applied a docking-guided *de novo* molecule generation workflow using CReM-opt and performed MD-based evaluation of identified hits.

Tubulin Inhibitor SAR Studies co-author of 4 publications

Performed molecular docking, molecular dynamics simulations, free energy calculations, ProLIF-based protein–ligand interaction analysis, parameterization of boron-containing molecules using Gaussian.

EasyDock [GitHub](#)

Implemented a Python-based AutoDock Vina workflow as part of a scalable molecular docking tool.

Selected Publications

9 total — [full list on ORCID](#)

- A. Ivanova, O. Mokshyna, and P. Polishchuk. "StreaMD: the toolkit for high-throughput molecular dynamics simulations". In: *Journal of Cheminformatics* (2024). DOI: 10.1186/s13321-024-00918-w
- A. Nikonenko et al. "Multiple Conformer Descriptors for QSAR Modeling". In: *Molecular Informatics* (2021). DOI: 10.1002/minf.202060030
- F. Li et al. (incl. Ivanova A.). "CACHE Challenge #1: Targeting the WDR Domain of LRRK2, A Parkinson's Disease Associated Protein". In: *Journal of Chemical Information and Modeling* (2024). DOI: 10.1021/acs.jcim.4c01267

Honors & Awards

2026	3rd Place (individual on-site entry) <i>in silico</i> Drug Design Challenge, Olomouc, Czech Republic Workshop "Advanced <i>in silico</i> Drug Design"
2022/2025	Dean's Award Palacký University Significant publication activity 2021/2024 — "Multiple Conformer descriptors for QSAR modeling", "StreaMD: the toolkit for high-throughput molecular dynamics simulations"
2023	3rd Place Team CACHE Challenge #1 Prediction and experimental validation of hits for WDR domain of LRRK2
2022	Poster Award 12th ICCS, Netherlands GenCReM: <i>de novo</i> molecular generation

Lectures & Workshops

2023–2026	Lecturer, annual workshop "Advanced <i>in silico</i> Drug Design" Palacký University, Olomouc, Czech Republic Molecular Dynamics Workshop
2025	Talk, RDKit UGM 2025 Prague, Czech Republic CReM-opt: <i>de novo</i> design guided by explainable docking